

GROUP ASSIGNMENT

BUDGETING FOR OPERATIONS IN MANUFACTURING FIRM

DTN Boat Company is a manufacturing company that makes fishing boat. The boats are made of fiberglass and wood and DTN only make one type – a 15-foot fishing boat. Subird actual balance sheet for 2023 is shown in the table 1 below:

DTN Boat Company			
Balance Sheet (Actual)			
December 31,2023			
Assets			
Current assets:			
Cash		\$ 137,000	
Accounts receivable		60,000	
Direct materials inventory ¹		8,000	
Finish goods inventory ²		<u>50,000</u>	\$ 255,000
Long-term assets:			
Land		\$ 231,500	
Building and equipment		1,500,000	
Less acumulated depreciation		<u>(120,000)</u>	<u>\$ 1,611,500</u>
Total assets			<u>\$1,866,500</u>
Liabilites and Stockholder's Equity			
Current liabilities:			
Account payable ³		\$ 10,000	
Income taxes payable ³		<u>96,000</u>	\$ 106,000
Stockholder's equity:			
Common stock, \$5 par		\$ 1,000,000	
Paid-in captial in excess of par		250,000	
Retained earnings		<u>510,500</u>	<u>1,706,500</u>
Total liabilities and stockholder's equity			<u>\$1,866,500</u>

Notes:

¹Composed of 600 board feet of lumber and 400 feet of fiberglass

²Composed of 10 finished boats.

³Expected to be paid in the first quarter of 2024

- DTN Boat company has projected 2024 sales to be 100 boats. The anticipated sales price for each boat is \$10,000.
- DTN Boat company projects its boat sales on a quarterly basis.
- The marketing department has effectively researched all the relevant variables in order to predict the number of boats that will be sold each quarter. The data of sales volume is show in table 2 below:

DTN Boat Company	
Sales Volume Estimation 2024	
Quarter 1	20 boats
Quarter 2	40 Boats
Quarter 3	30 Boats
Quarter 4	10 Boats

- Based on past experience, DTN is able to collect approximately 80% of revenue from customers in the quarter in which the sale took place, with the remaining 20% collected in the following quarter.
- DTN Boat Company has determined that its desired ending inventory for each quarter should be approximately 50% of projected sales for the next quarter. The fourth quarter's ending inventory is 15 boats, which is 50% of the next quarter's (first quarter of 2025) expected sales.
- DTN's engineers estimate that the standard amounts of wood and fiberglass needed per boat are as follows:

15-Foot Boat

Direct materials requirements:

- | | |
|--------------|----------------|
| - Wood | 100 board feet |
| - Fiberglass | 40 square feet |
- DTN's purchasing agent has been able to purchase wood at \$10 per board foot and fiberglass at 5\$ per square foot. The purchasing agent expects that these prices will continue to be avaiable through 2024.

- DTN's management has decided to maintain a supply of 30% and 50% of the next quarter's production requirements for wood and fiberglass, respectively.
- DTN does not pay for raw materials at the time of the purchase, which means that DTN maintains an accounts payable balance.
- DTN's policy is to pay for 75% of its raw materials purchases in the current quarter, and the remaining 25% in the subsequent quarter.
- The direct labor budget for DTN is based on an average hourly wage rate of \$20 per hour for production workers and 80 labor hours to make a 15-foot boat. The \$20 per-hour wage rate includes fringe benefits (i.e., the cost of sick leave, vacation pay, insurance, etc.) and payroll taxes. Assume that DTN's budgeted labor costs for each quarter are all paid in that same quarter.
- Assume that volume of direct labor hour is a good predictor of variable manufacturing overhead cost for DTN, and that DTN uses direct labor hours to allocate these costs to each quarter.
- Total fixed costs are simply allocated evenly across the four quarters. The information about variable and fixed MOH is listed in the table below:

Variable MOH Costs	
Indirect materials	\$1.5
Indirect labor	5
Utilities	1
Fixed MOH Costs	
Property tax	\$ 2000
Insurance	1000
Depreciation – plant	15,000
Supervisor's salaries	24,000

- Assume total estimated delivery expense for 2024 is \$50,000. Sales Commission fee is 15% of the selling price each boat. The information of fixed selling and administrative expense is as follows:

Fixed S&A Expenses	
Executives' salaries	\$ 34,000
Depreciation expense	5,000
Advertising expense	10,000
Miscellaneous expense	2,500

- DTN Boat Company desired to maintain at least \$100,000 cash balance at the end of each quarter. The interest rate of 2024 is 10%. DTN plans to declare dividends of \$30,000 in the first quarter of

2024, which is expected to pay in the third quarter. DTN intends to purchase an equipment at the end of 2024.

Requirement:

Task 1: Prepare the sales budget for DTN Boat Company following the form below:

DTN Boat Company 2024 Sales Budget					
	Q1	Q2	Q3	Q4	Total
Sales volume	20	40	30	10	100
Price	10,000	10,000	10,000	10,000	
Revenue earned (Sales volume* Price)	200,000	400,000	300,000	100,000	1,000,000
Received in current quarter (Rev x 80%)	160,000	320,000	240,000	80,000	800,000
Prior revenue collected	60,000	40,000	80,000	60,000	240,000
Collections from customers	220,000	360,000	320,000	140,000	1,040,000

Received in current quarter= Revenue earned*80% (80% of revenue from customers in the quarter in which the sale took place) 80% Collected Current Quarter Là 80% doanh thu của chính quý đó.

Prior revenue collected= Revenue earned của quý trước*20%

- Q1 thu khoản phải thu cuối năm 2023. Balance Sheet 31/12/2023: AR = 60,000 => Prior Revenue Collected Q1 = 60,000

- Q2= Rev earned Q1*20%= 200,000*20% =40,000

- Q3= Rev earned Q2*20%= 400,000*20% =80,000

- Q4= Rev earned Q3*20%= 300,000*20% =60,000

Collections from customers = Received in current quarter + Prior revenue collected

Task 2: prepare production budget for DTN Boat Company following the form below:

DTN Boat Company 2024 Production Budget					
	Q1	Q2	Q3	Q4	Total
Sales volume (Task 1)	20	40	30	10	100
Desired ending inventory	20	15	5	15 (Đề cho)	55

	(50%*SaleQ2)	(50%*SaleQ3)	(50%*SaleQ4)		
Total boats needs	40	55	35	25	155
Less beginning inventory	10	20	15	5	50
Total boats to produce	30	35	20	20	105

Desired ending inventory= 50%* Sales Next Quarter (50% of the next quarter's)

- Q1= 50%*Sales Q2= 50%*40=20

- Q2= 50%*Sales Q3= 50%*30=15

- Q3= 50%*Sales Q4= 50%*10=5

- Q4= 15 (The fourth quarter's ending inventory is 15 boats, which is 50% of the next quarter's)

Total boats needs = Sales volume + Desired ending inventory

Less beginning inventory

Theo Balance Sheet: Finished goods inventory = 10 finished boats (Composed of 10 finished boats)

=> Đầu năm 2024, Q1 đã có: 10 boats trong kho => Less Beginning Inventory Q1 = 10

Lấy tồn kho cuối Q1 chuyển sang. Ở Q1: 50%×Sales Q2 = 50%×40=20 ⇒ Less Beginning Inventory Q2 = 20

Less Beginning Inventory Q3= 50%×30= 15

Less Beginning Inventory Q4= 50%×10= 5

Total boats to produce= Sales volume + Desired ending inventory - Less beginning inventory

Task 3: prepare Direct Materials budget for DTN Boat Company following the form below:

DTN Boat Company					
2024 Direct Materials Budget					
Wood	Q1	Q2	Q3	Q4	Total
Production volume (Task 2)	30	35	20	20	105
Standard feet per boat (*100 feet)	100	100	100	100	
Wood needed in production (Production volume* Standard feet per boat)	3,000 (30*100)	3,500	2,000	2,000	10,500
Desired ending inventory	1,050	600	600	900	3,150

	(30%*3,500)				
Total wood need (Wood needed in production + Desired ending inventory)	4,050 (3,000 +1,050)	4,100	2,600	2,900	13,650
Less beginning inventory	600	1,050	600	600	2,850
Total wood purchase (Total wood need - Less beginning inventory)	3,450 (4,050 – 600)	3,050	2,000	2,300	10,800
Standard purchase price	10	10	10	10	
Total wood purchases	34,500	30,500	20,000	23,000	108,000
Fiberglass	Q1	Q2	Q3	Q4	Total
Production volume (Task 2)	30	35	20	20	105
Standard feet per boat	40	40	40	40	
Fiberglass needed in production	1,200 (30*40)	1,400	800	800	4,200
Desired ending inventory	700	400	400	600	2,100
Total fiberglass need	1,900	1,800	1,200	1,400	6,300
Less beginning inventory	400	700	400	400	1,900
Total fiberglass purchase	1,500	1,100	800	1,000	4,400
Standard purchase price	5	5	5	5	
Total fiberglass purchases	7,500	5,500	4,000	5,000	22,000
	Q1	Q2	Q3	Q4	Total
Total materials purchases (total wood + fiberglass)	42,000 34,500+7,500	36,000	24,000	28,000	130,000
Paid in current quarter	75%	75%	75%	75%	
Current purchase payments	31,500	27,000	18,000	21,000	97,500
Prior purchase payments	10,000	10,500	9,000	6,000	35,500
Payments to suppliers	41,500	37,500	27,000	27,000	133,000

Production volume Wood = Total boats to produce (Task 2)

Desired ending inventory = 30%×Wood needed next quarter (DTN's management has decided to maintain a supply of 30% of the next quarter's production requirements for wood)

- Q1= Wood needed Q2*30%=> Tồn kho cuối Q1: $3500 \times 30\% = 1050$
- Q2= Wood needed Q3*30%=> Tồn kho cuối Q2: $2000 \times 30\% = 600$
- Q3= Wood needed Q4*30%=> Tồn kho cuối Q3: $2000 \times 30\% = 600$
- Q4 15 boats, which is 50% of the next quarter's (first quarter of 2025) expected sales => $15 = 50\% \times \text{Sales Q1/2025}$
 $\text{Sales Q1/2025} \Rightarrow \text{Sales Q1/2025} = 15 \div 50\% = 30 \text{ boats}$
 $30 \text{ boats} \times 100 = 3,000 \text{ board feet} \Rightarrow \text{Q4} = 30\% \times 3,000 = 900$

Total wood need= Wood needed in production + Desired ending inventory

Less beginning inventory:

Balance Sheet cho: Raw Materials Inventory: Wood = 600 board feet => Q1= 600 board feet

Q2, tồn kho đầu kỳ là tồn kho cuối Q1: 1050 => Q2= 1,050 board feet

Q3, tồn kho đầu kỳ là tồn kho cuối Q2: => Q3= 600 board feet

Q4, tồn kho đầu kỳ là tồn kho cuối Q3: 1050 => Q4= 600 board feet

Total wood purchase= Total wood need - Less beginning inventory

Standard purchase price = 10 (DTN's purchasing agent has been able to purchase wood at \$10 per board foot)

Total wood purchases= Standard purchase price* Total wood purchase

Production volume Fiberglass = Total boats to produce (Task 2)

Fiberglass needed in production = Production volume Fiberglass* Standard feet per boat

Desired ending inventory = 50%× Fiberglass needed next quarter

Q1= 50%* Fiberglass needed Q2= 50%* 1400=700

Q2= 50%* Fiberglass needed Q3= 50%* 800= 400

Q3= 50%* Fiberglass needed Q4= 50%* 800= 400

Q4: Sales Q1/2025=30 boats => $30 \times 40 = 1200 \Rightarrow 1200 \times 50\% = 600$

Total fiberglass need = Fiberglass needed in production + Desired ending inventory

Less beginning inventory:

Balance Sheet đầu năm có: Raw Materials Inventory= 400 square feet of fiberglass

=> Beginning Inventory Q1=400

Đầu Q2 là cuối Q1=> Beginning Inventory Q2=700

Đầu Q3 là cuối Q2=> Beginning Inventory Q3=400

Đầu Q4 là cuối Q3=> Beginning Inventory Q4=400

Total fiberglass purchase= Total fiberglass need - Less beginning inventory

Total fiberglass purchases= Total fiberglass purchase* Standard purchase price

Total materials purchases= Wood purchases + Fiberglass purchases

Paid in current quarter = 75% (DTN 's policy is to pay for 75% of its raw materials purchases in the

current quarter, and the remaining 25% in the subsequent quarter)

Current purchase payments = Paid in current quarter* Total materials purchases

Prior purchase payments

Q1= AP= 10000

Q2=42000*25%=10500

Q3=36000*25%=9000

Q4=24000*25%=6000

Payments to suppliers= Current purchase payments + Prior purchase payments

Task 4: prepare Direct Labor Budget for DTN Boat Company following the form below:

DTN Boat Company					
2024 Direct Labor Budget					
	Q1	Q2	Q3	Q4	Total
Production volume (Task 2)	30	35	20	20	105
Standard hours per boat (Đề cho)	80	80	80	80	
Budgeted labor hours = Production volume × Standard hours per boat	2,400 30*80	= 2,800 35*80	= 1,600 20*80	= 1,600 20*80	= 8,400
Standard wage rate per hour (Đề cho)	20	20	20	20	
Direct labor cost = Budgeted labor hours × Standard wage rate	48,000 2,400*20	= 56,000 2,800*20	= 32,000 1,600*20	= 32,000 1,600*20	= 168,000

- Standard hours per boat: DTN requires 80 direct labor hours to produce one 15-foot boat.
- Standard wage rate per hour: Average hourly wage rate = \$20 per hour.

Task 5: prepare Manufacturing Overhead Budget for DTN Boat Company following the form below:

Ⓢ Assumption: Fixed MOH is the cost for one year

DTN Boat Company						
2024 Manufacturing Overhead Budget						
Variable MOH costs		Q1	Q2	Q3	Q4	Total
Total labor hour (Task 4)		2,400	2,800	1,600	1,600	8,400
Variable MOH rates per hour: (Đề \$		7.5	7.5	7.5	7.5	7.5

cho)						
Indirect materials = Total labor hours × \$1.5	1.5	3,600 = 2,400*1.5	4,200 = 2,800*1.5	2,400 = 1,600*1.5	2,400 = 1,600*1.5	12,600
Indirect labor = Total labor hours × \$5	5	12,000 = 2400*5	14,000 = 2800*5	8,000 = 1600*5	8,000 = 1600*5	42,000
Utilities = Total labor hours × \$1	1	2,400 = 2,400*1	2,800 = 2,800*1	1,600 = 1,600*1	1,600 = 1,600*1	8,400
Total variable MOH costs = Indirect Materials + Indirect Labor + Utilities	7.5	18,000	21,000	12,000	12,000	63,000
Fixed MOH cost per quarter		Q1	Q2	Q3	Q4	Total
Property tax	2,000	500 = 2,000/4	500	500	500	2,000
Insurance	1,000	250 = 1,000/4	250	250	250	1,000
Depreciation – plant	15,000	3,750 = 15,000/4	3,750	3,750	3,750	15,000
Supervisor's salaries	24,000	6,000 = 24,000/4	6,000	6,000	6,000	24,000
Total fixed MOH costs	42,000	10,500 = 42,000/4	10,500	10,500	10,500	42,000
Total		Q1	Q2	Q3	Q4	Total
Total MOH costs = Variable MOH + Fixed MOH		28,500 = 18,000 + 10,500	31,500 = 21,000 + 10,500	22,500 = 12,000 + 10,500	22,500 = 12,000 + 10,500	105,000
Less depreciation		(3,750)	(3,750)	(3,750)	(3,750)	(15,000)
Total MOH payments = Total MOH Costs – Depreciation		24,750 = 28,500 – 3,750	27,750 = 31,500 – 3,750	18,750 = 22,500 – 3,750	18,750 = 22,500 – 3,750	90,000

- Variable MOH rates per hour:

Variable MOH Costs	
Indirect materials	\$1.5
Indirect labor	5
Utilities	1

Tổng Variable MOH Rate = $1.5 + 5 + 1 = 7.5$ USD/hour

- Fixed MOH cost per quarter

Fixed MOH Costs	
Property tax	\$ 2000
Insurance	1000
Depreciation – plant	15,000
Supervisor's salaries	24,000

Tổng Fixed MOH: = $2,000 + 1,000 + 15,000 + 24,000 = 42,000$ USD

Task 6: prepare Annual Budgeted Production Cost Sheet for DTN Boat Company following the form below:

DTN Boat Company 2024 Budgeted Product Cost Sheet		
	\$	Total
Direct materials cost per boat (Đề cho)		
- Wood	10	$1,000 = 100 \text{ board feet} \times 10$
- Fiberglass	5	$200 = 40 \text{ square feet} \times 5$
Direct labor cost per boat (Đề cho)	20	$1,600 = 80 \times 20$
Variable MOH cost per boat (Task 5)	7.5	$600 = 80 \times 7.5$
Total variable cost per boat = Direct Materials + Direct Labor + Variable MOH	42.5	$3,400 = 1,000 + 200 + 1,600 + 600$
Total fixed MOH (Task 5)		42,000
Divided by production volume (Task 2)		105
Fixed MOH cost allocated per boat = Total Fixed MOH ÷		$400 = 42,000 / 105$

Production Volume		
Total production cost per boat = Total Variable Cost per Boat + Fixed MOH per Boat		3,800 = 3,400 + 400

Task 7: prepare Selling and Administrative (S&A) Expense Budget for DTN Boat Company following the form below:

à Assumption: Fixed S&A Expenses is the cost for one year

DTN Boat Company						
2024 Selling and Administrative (S&A) Expense Budget						
Variable S&A expenses		Q1	Q2	Q3	Q4	Total
Sales volume		20	40	30	10	100
Variable expense per boat sold:	\$					
- Delivery expense	500 (\$50,000/100)	10,000 (20*500)	20,000 (40*500)	15,000 (30*500)	5,000 (10*500)	50,000
- Sales commissions	1,500 (15%*10,000)	30,000 (20*1,500)	60,000 (40*1,500)	45,000 (30*1,500)	15,000 (10*1,500)	150,000
Total variable S&A expenses	2,000	40,000	80,000	60,000	20,000	200,000
Fixed S&A expenses per quarter	\$	Q1	Q2	Q3	Q4	Total
Executives' salaries	8,500 (\$34,000/4)	8,500	8,500	8,500	8,500	34,000
Depreciation expense	1,250 (\$5,000/4)	1,250	1,250	1,250	1,250	5,000
Advertising expense	2,500 (\$10,000/4)	2,500	2,500	2,500	2,500	10,000
Miscellaneous expense	625 (\$2,500/4)	625	625	625	625	2,500
Total fixed S&A expenses	12,875	12,875	12,875	12,875	12,875	51,500
Total		Q1	Q2	Q3	Q4	Total

Total S&A expenses		52,875 (40,000 + 12,875)	92,875 (80,000 + 12,875)	72,875 (60,000 + 12,875)	32,875 (20,000 + 12,875)	251,500
Less depreciation		(1,250)	(1,250)	(1,250)	(1,250)	(5,000)
Total S&A payments		51,625	91,625	71,625	31,625	246,500

à Total S&A expenses = Total variable S&A expenses + Total fixed S&A expenses

Task 8: prepare Cash Budget for DTN Boat Company following the form below:

- DTN intends to purchase an equipment at the end of 2024 à Based on Accumulated Depreciation, calculate the equipment the business intends to purchase as follows:

- Giả định: DN hướng đến nguyên tắc Đầu tư bù đắp hao mòn (Replacement Capital Expenditure), mục tiêu của chính sách này là giữ cho Giá trị còn lại (Net Book Value - NBV) của tài sản cố định không đổi sau một năm hoạt động:

Accumulated depreciation 2024 (BS 2023, task 5, task 7): $120,000 + 5,000 + 15,000 = 140,000$

- Building and equipment: \$1,500,000

- Accumulated depreciation: \$120,000

à NBV beginning 2024: $\$1,500,000 - \$120,000 = \$1,380,000$

To maintain production capacity, the ending NBV must equal the beginning NBV of \$1,380,000. Therefore, we have the following equation:

Ending NBV = (Beginning Gross Assets + New Equipment) - Ending Accumulated Depreciation

=> $\$1,380,000 = (\$1,500,000 + \text{New Equipment}) - \$140,000$

=> New Equipment purchase = \$20,000

DTN Boat Company					
2024 Cash Budget					
	Q1	Q2	Q3	Q4	Total
Receipts:					

- Collections from customers (Task 1)	220,000	360,000	320,000	140,000	1,040,000
Payments:					
- Direct materials (Task 3)	41,500	37,500	27,000	27,000	133,000
- Direct labor (Task 4)	48,000	56,000	32,000	32,000	168,000
- Manufacturing Overhead (Task 5)	24,750	27,750	18,750	18,750	90,000
- Selling and Administrative expenses (Task 7)	51,625	91,625	71,625	31,625	246,500
- Income tax payment for 2023 (Balance Sheet)	96,000	0	0	0	96,000
- Interest payment (10%)	0	122	0	0	122
- Dividends (Đề cho)	0	0	30,000	0	30,000
- Equipment purchase (calculated)	0	0	0	20,000	50,000
Total payments	261,875	212,997	179,375	129,375	783,622
Financing:					
- Beginning cash balance	137,000	100,000	242,128	382,753	
- Total receipts	220,000	360,000	320,000	140,000	1,040,000
- Total payments	(261,875)	(212,997)	(179,375)	(129,375)	(783,622)
Cash available	95,125	247,003	382,753	393,378	
- Minimum cash balance desired	(100,000)	(100,000)	(100,000)	(100,000)	(400,000)
- Cash excess (deficiency)	(4,875) = 95,125 - 100	147,003 247 - 100	282,753 382,753 -100	293,378 393,378 - 100	
- Borrowings	4,875	0	0	0	4,875
- Repayments of principle	0	4,875	0	0	4,875
Total financing inflows (outflows)	4,875	(4,875)	0	0	
Ending cash balance	100,000	242,128	382,753	393,378	

Beginning cash balance:

- Q1: 137,000 (Balance Sheet)

- Q2: $137,000 + 220,000 - 261,875 = 95,125$ (DTN Boat Company desired to maintain at least \$100,000 cash balance at the end of each quarter) à 100,000.

DN vay thêm 4,875 để có được tồn quỹ 100,000; lãi vay phải trả trong Q2: $4,875 \times 2.5\% = 121.875 \sim 122$.

- Q3: $100,000 + 360,000 - 212,997 = 247,003$; hơn 147,003 so với mức tối thiểu à trả nợ vay 4,875 còn 242,128

- Q4: $242,128 + 320,000 - 129,375 = 393,378$

Ending cash balance:

- Q1: 100,000 (do mức tiền mặt tối thiểu)

- Q2: là begin của Q3 = 242,128 đã tính

- Q3: là begin của Q4 = 382,753

- Q4: 393,378

Task 9: prepare Budgeted Cost of Goods Sold for DTN Boat Company following the form below:

DTN Boat Company	
Budgeted Cost of Goods Sold	
For the year Ended December 31, 2024	
Direct materials purchases (task 3)	130,000
Direct materials beginning inventory (Balance sheet 2023)	8,000
Less direct materials ending inventory (task 3)	12,000 (900*10+600*5)
Direct materials costs in production	126,000 (130,000+8,000-12,000)
Direct labor costs in production (task 4)	168,000
MOH costs in production (task 5)	105,000
- Cost of goods manufactured	399,000 (126,000+168,000+105,000)
Finished goods beginning inventory (Balance sheet 2023)	50,000

Less finish goods ending inventory (task 2, task 6)	57,000 (15*3,800)
- Cost of goods sold	392,000 (399,000+50,000-57,000)

Direct materials ending inventory = Ending inventory (Wood) * Purchase price (Wood)
+ Ending inventory (Fiberglass)* Purchase price(Fiberglass)

Direct materials costs in production = Direct materials purchases + Direct materials beginning inventory

- Direct materials ending inventory

Cost of goods manufactured = Direct materials costs in production + Direct labor costs in production
+ MOH costs in production

Finish goods ending inventory = Ending inventory (Q4) * Total production cost per boat

Cost of goods sold = Cost of goods manufactured + Finished goods beginning inventory

- Finished goods ending inventory

Task 10: prepare Budgeted Income Statment for DTN Boat Company following the form below:

DTN Boat Company Budgeted Income Statement For the year Ended December 31, 2024	
Sales revenue	1,000,000
Cost of goods sold (task 9)	(392,000)
- Gross margin	608,000
Selling and Administrative expense (task 7)	251,500
- Operating income	356,500
Interest expense (task 8)	(122)
Net income before tax	356,378
Income taxes (40%)	142,551.2
Net income after tax	213,826.8

Sale revenue = 100 boats* 10000\$

Gross margin = Sale revenue – Cost of goods sold

Operating income= Gross margin - Selling and Administrative expense

Net income before tax= Operating income – Interest Expense

Income taxes (40%) = Net income before tax*40%

Net income after tax = Net income before tax - Income taxes (40%)

Task 11: prepare Budgeted Balance Sheet for DTN Boat Company following the form below:

DTN Boat Company Budgeted Balance Sheet December 31,2024		
Assets		
Current assets:		
Cash (task 8)	393,378	
Accounts receivable (task 1)	20,000 (10,000*20%)	
Direct materials inventory ¹ (task 9)	12,000	
Finish goods inventory ² (task 9)	<u>57,000</u>	482,378
Long-term assets:		
Land (BS 2023)	231,500	
Building and equipment (task 8)	1,520,000 (1,500,000+20,000)	
Less accumulated depreciation (BS 2023, task 5, task 7)	<u>(140,000)</u> (120,000+15,000+5,000)	<u>1,611,500</u>
Total assets		<u>2,093,878</u>
Liabilites and Stockholder's Equity		
Current liabilities:		
Account payable ³ (Task 3)	7,000 (28,000*25%)	
Income taxes payable ³ (Task 10)	142,551.2	149,551.2
Stockholder's equity:		

Common stock, \$5 par	1,000,000	
Paid-in capital in excess of par	250,000	
Retained earnings	<u>694,326.8</u> (510,500+213,826.8-30,000)	<u>1,944,326.8</u>
Total liabilities and stockholder's equity		<u>2,093,878</u>

Accounts receivable = Revenue earned (Q4) * 20% (80% of revenue from customers in the quarter in which the sale took place, with the remaining 20% collected in the following quarter)

Building and equipment = Building and equipment (BS 2023) + Equipment purchase

Less accumulated depreciation = Less accumulated depreciation (BS 2023) + Depreciation plant
+ Depreciation expense

Account payable = Total materials purchases (Q4) * 25% (DTN 's policy is to pay for 75% of its raw materials purchases in the current quarter, and the remaining 25% in the subsequent quarter)

Retained earnings = Beginning RE + Net income after tax – Dividends (DTN plans to declare dividends of \$30,000 in the first quarter of 2024)

Task 12: prepare Budgeted Statement of Cash Flows for DTN Boat Company following the form below:

DTN Boat Company		
Budgeted Income Statement		
For the year Ended December 31, 2024		
Cash flow from operating activities:		
Net income (task 10)	213,826.8	
Adjustment:		
- Depreciation	20,000	
- Accounts receivable decrease	40,000	
- Direct materials inventory increase	(4,000)	
- Finished goods inventory increase	(25,000)	
- Account payable decrease	(3,000)	
- Income taxes payable increase	46,551.2	

Net cash provided by operating activities		<u>288,378</u>
Cash flow from investment activities		
- Purchase of equipment (task 8)	(20,000)	
Net cash used in investment activities		<u>(20,000)</u>
Cash flow from financing activities		
- Cash obtained from borrowing (task 8)	4,875	
- Payment of dividends (task 11)	(30,000)	
- Repay of borrowing (task 8)	(4,875)	
Net cash used in financing activities:		<u>(30,000)</u>
- Net increase in cash		238,378
Beginning cash balance		<u>137,000</u>
Ending cash balance		<u>375,378</u>

Adjustment:

Depreciation:

+ Plant depreciation (Task 5): 15,000

+ S&A depreciation (Task 7): 5,000

Account Receivable:

+ Account Receivable (Balance sheet 2023): 60,000

+ Account Receivable (Task 11): 20,000

⊖ **Account Receivable decrease : 40,000**

Direct materials inventory:

+ Beginning Direct materials inventory: 8,000

+ Ending Direct materials inventory: 12,000

⊖ **Direct materials inventory increase: 4,000**

Finished goods inventory:

+ Beginning Finished goods inventory: 50,000

+ Ending Finished goods inventory: 75,000

=> Finished goods inventory increase: 25,000

Account payable :

+ Beginning Account payable : 10,000

+ Ending Account payable : 7,000

⊖ Account payable decrease: 3000

Income taxes payable:

+ Beginning Income taxes payable: 96,000

+ Ending Income taxes payable: 142,551.2

⊖ Income taxes payable increase: 46,551.2

Purchase of equipment: “ DTN intends to purchase an equipment at the end of 2024 à 20,000\$.”

Payment of dividends: “DTN plans to declare dividends of \$30,000 in the first quarter of 2024”

Cash obtained from borrowing:

In the first quarter, the company needed additional cash to maintain the minimum required cash balance of \$100,000; Cash available => DTN borrowed: 4,875

In the second quarter, the company had sufficient cash and repaid the loan.